

Esteban Mares

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EXPERIENCE

Software Development Engineer Intern

June 2024 – September 2024

Amazon Web Services

Seattle, WA

- Identified JSON serialization overhead as the root cause of timeout failures in a KDS caching service processing millions of events per second.
- Migrated the service to Amazon Ion binary serialization, reducing P99 latency by 40% using a schema-less binary format to handle variable-shaped objects.
- Authored a 5,000+ word Technical Design Document (TDD) covering problem analysis, protocol evaluation, and test plan, delivering two weeks ahead of schedule.
- Validated the migration end-to-end by writing new JUnit unit and integration tests for the Ion serialization layer, maintaining system stability throughout deployments.

Inventory Analyst

November 2021 – June 2024

Fleetpride

Visalia, CA

- Automated data validation using Python (Pandas) and Excel Macros, replacing manual verification and saving 10+ hours/week of labor.
- Built ETL workflows to clean supply chain datasets, improving data accuracy from 85% to 96% and preventing \$165K+ in annual losses.
- Analyzed \$5M in inventory to identify root causes of inefficiencies, reducing holding costs by 17%.

PROJECTS

UFC Fight Prediction System | *React, TypeScript, Python, FastAPI, PostgreSQL, LangGraph*

- Built a four-node LangGraph agent pipeline that coordinates live odds fetching, ML predictions, and Claude-powered fight analysis.
- Trained a calibrated Logistic Regression model on 4,768 fights, achieving 62.7% accuracy and 0.642 AUC-ROC across 15 leak-free features.
- Added a betting layer with calibrated probabilities, EV thresholding at 5% minimum edge, and Quarter Kelly sizing.

Label Generation Platform | *React, Python, PostgreSQL, Docker*

- Built a full-stack label generation platform with a multithreaded Python backend, processing 50,000+ SKUs and reducing annual inventory from 5 days to 2.
- Built an Excel ingestion pipeline to parse warehouse exports into the label generation engine.
- Deployed with Docker, JWT authentication, and rate limiting. Actively used in production.

Vehicle Value Prediction System | *Python, Flask, Scikit-learn, PostgreSQL*

- Trained a Random Forest model on 3M+ vehicle records, achieving an R^2 of 0.829 predicting market values within \$1,950 on average.
- Built an ETL pipeline with NumPy and Pandas to process raw data for a React frontend visualizing predicted values and depreciation timelines.

TECHNICAL SKILLS

Languages: Java, Python (Pandas/Flask), SQL, TypeScript, JavaScript

Backend & Data: Spring Boot, Node.js, Microservices, REST APIs, PostgreSQL, Kinesis, Multithreading

Cloud & DevOps: AWS (EC2, Lambda, S3, Kinesis, CloudWatch), Docker, CI/CD (GitHub Actions), Terraform, Git

Frontend: React.js, HTML5, CSS3, Tailwind CSS

Concepts: Distributed Systems, System Design, Agile, ETL, OOP, Data Structures & Algorithms, Linux

EDUCATION

Western Governors University

Bachelor of Science in Computer Science

November 2025